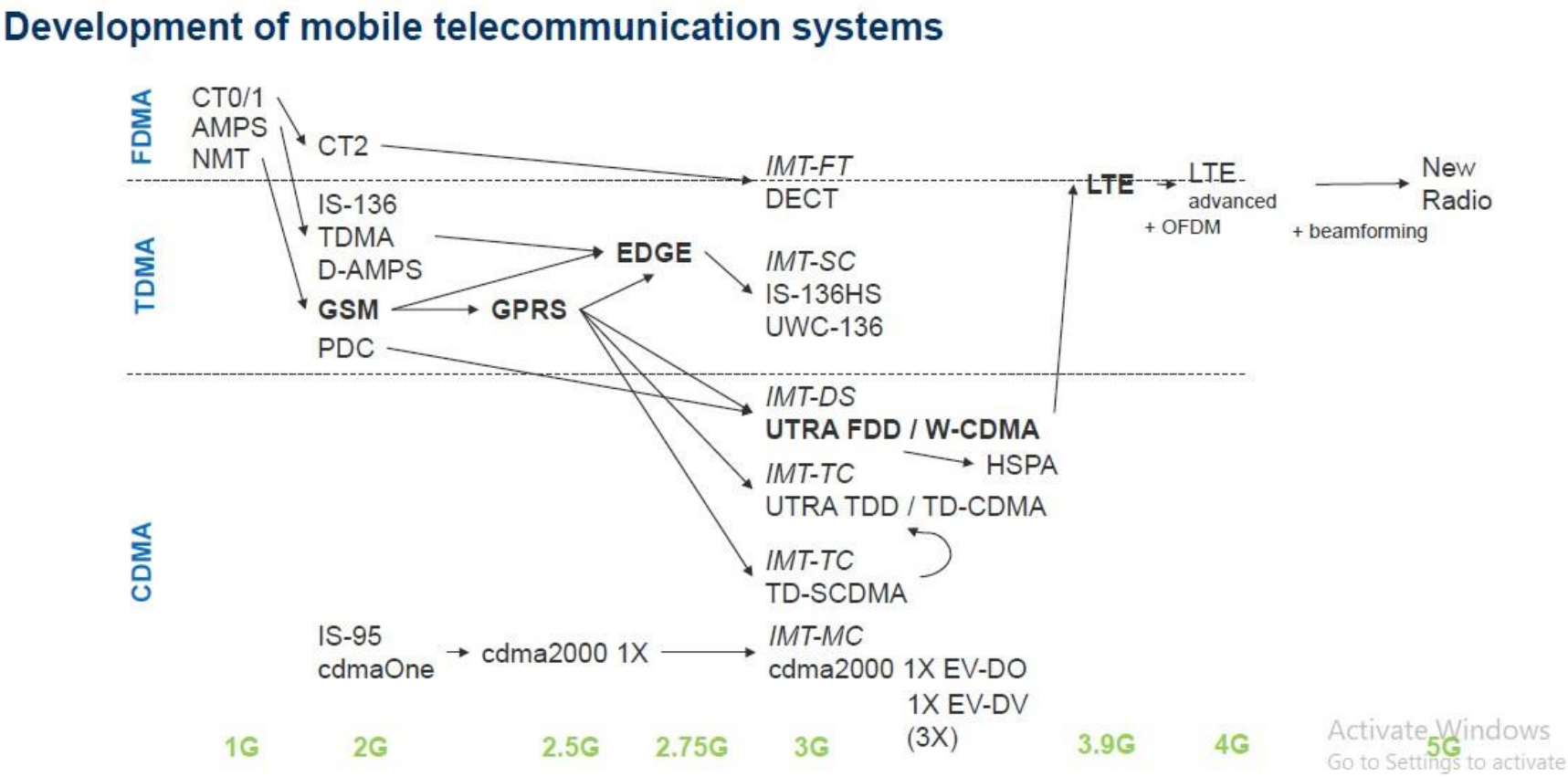
Unit II: TELECOMMUNICATION SYSTEMS



Bearer Services:

provide the **basic transmission capability** for moving signals between two points.

They do **not** define the content (voice, video, data) — only the transport.

Key Features:

- Provide data transmission over network
- Define speed, bandwidth, and quality of service
- Transparent to the type of information being carried

Examples:

- Internet data connection
- Leased lines
- ISDN bearer channels
- Mobile data (3G/4G/5G transport layer)

 Think of bearer service as the **"road"** that carries vehicles (information).

TelematicServices:
combine **telecommunications + information processing**.
They allow users to **send, receive, and process information**.
Key Features:

- End-to-end communication services
- Include data processing functions
- Built on top of bearer services

Examples:

- Email
- Fax
- Video conferencing
- Online banking
- Telemedicine

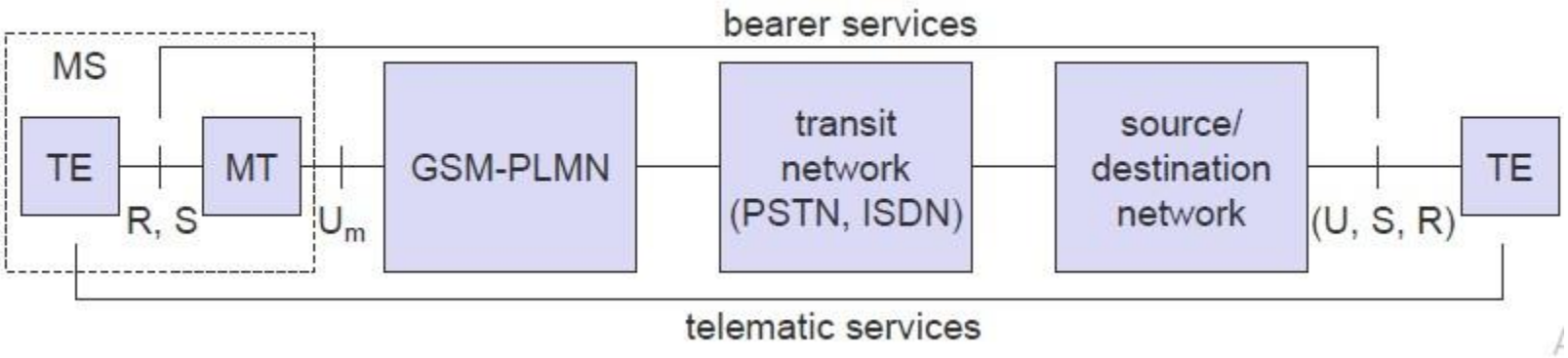
 Telematic service is the **"vehicle"** traveling on the road.
Supplementary Services:
additional features that enhance basic telecommunication services.
They do not work alone — they support bearer or telematic services.
Key Features:

- Improve convenience
- Add control or extra functionality
- Work alongside basic services

Examples:

- Caller ID
- Call waiting
- Call forwarding
- Voicemail
- Call barring

 Supplementary services are like **extra features in a car** (GPS, music system, etc.).



MS (Mobile Station)

The Mobile Station consists of:

- **TE (Terminal Equipment)** → The actual user device
  - Example: laptop, fax machine, phone handset

- **MT (Mobile Termination)** → The GSM radio part
  - Handles wireless communication with the network

So **MS = TE + MT**

Um Interface

- The **radio interface** between the mobile device (MT) and the GSM network. ● This is the wireless air link.

GSM-PLMN

- GSM Public Land Mobile Network
- The mobile operator's network (switching, routing, mobility management).

Transit Network (PSTN / ISDN)

- External networks that carry communication further. ● Examples:
    - PSTN (Public Switched Telephone Network)
    - ISDN (Integrated Services Digital Network)

Source/Destination Network

- The final network where the receiving user is located.
- Could be another GSM network, PSTN, ISDN, etc.

2 Bearer Services (Top Path in Diagram)

The top line labeled **"bearer services"** shows:

MS → GSM-PLMN → Transit Network → Destination

This path provides:

- Pure transmission capability
- Transport of data/voice
- No processing of content

It only ensures the signal is carried from one end to another.

Think of it as the **transport pipeline**.

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3 Telematic Services (Bottom Bracket)

The bottom bracket labeled **"telematic services"** spans from TE to TE.

This means:

Communication happens end-to-end between terminal devices.

Telematic services include:

- Fax
- Email
- Video calls
- Data communication

These services:

- Use bearer services
- Include processing and formatting of information

So telematic services sit **on top of bearer services**.

4⃣ What Do R, S, U Mean?

These indicate different types of services:

- **R** → Relay services
- **S** → Supplementary services
- **U** → User services

They classify how the service behaves across interfaces.

- GSM, GPRS, TETRA, UMTS, IMT-2000, LTE

Technology	Generation		Main Use	Data Rate
GSM (Global System for Mobile Communications)	2G	Uses **TDMA + FDMA**	Voice, SMS	9.6 kbps

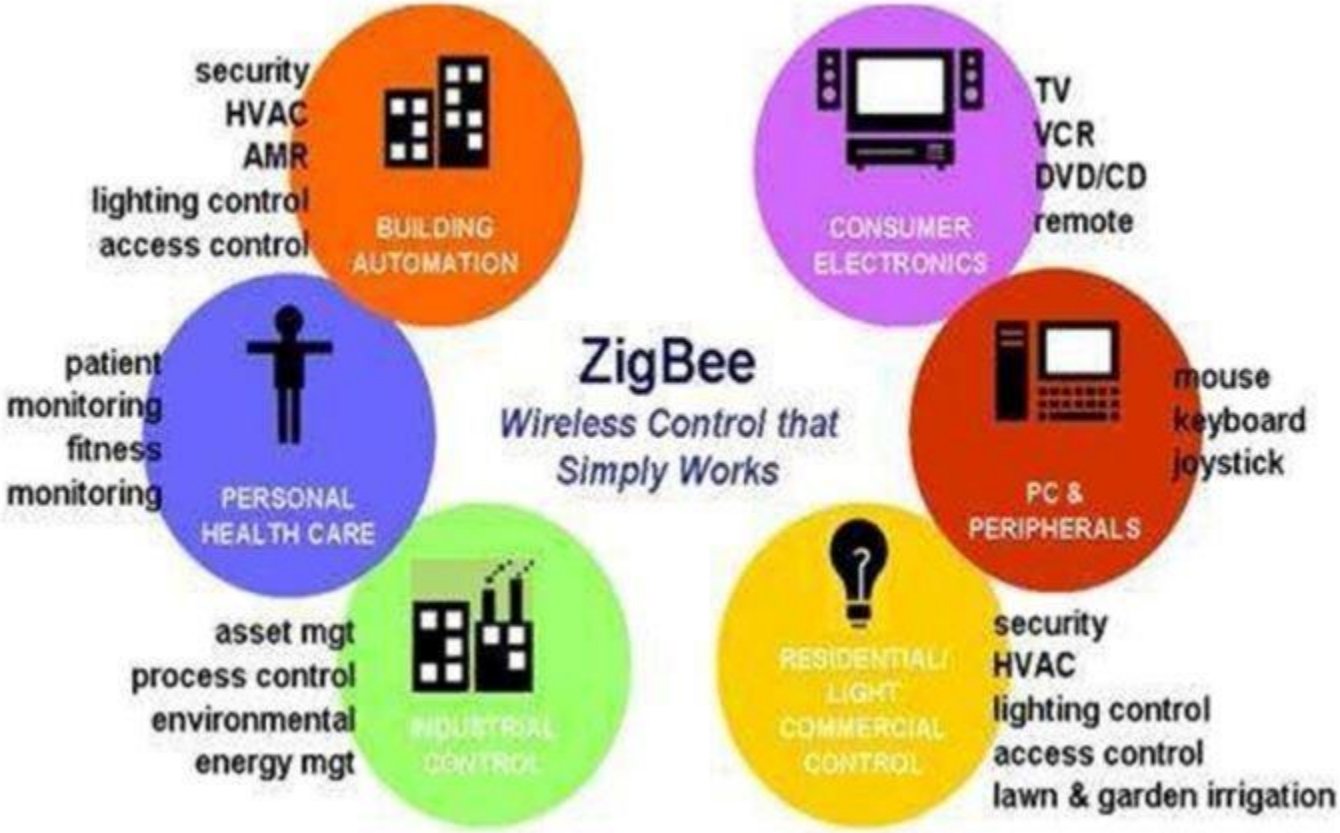
GPRS(General Packet Radio Service)	2.5G	Packet switching instead of circuit switching (dedicated circuit throughout the session - telephone)	Packet data (mail, audio/video pkts)	~100 kbps
TETRA (Terrestrial Trunked Radio)	Special	Group Call, Push to Talk	Public safety	Voice + data (Police, Fire services, Military
				Transport, authorities)
UMTS (Universal Mobile Telecommunications System)	3G	WCDMA	Multimedia	up to 2 Mbps

IMT-2000 (International Mobile Telecommunications-2000)	3G framework	Not a single technology, UMTS (WCDMA), CDMA2000, TD-SCDMA	Global standard	—
LTE (Long Term Evolution)	4G	OFDMA (downlink) and SC-FDMA (uplink)	Mobile broadband	100+ Mbps (HD video, VoIP, gaming)

Wireless LAN (WLAN)s/Personal Area Network(PAN)s

Feature	**WLAN (Wi-Fi)**	**WPAN (Bluetooth/ZigBee)**
Range	10–100 m	1–10 m
Power	Medium–High	Very low
Data rate	High	Low
Devices	PCs, phones	Sensors, wearables
Standard	IEEE 802.11	IEEE 802.15

Bluetooth: Medium speed, device connectivity
ZigBee: Ultra-low power, sensor networks



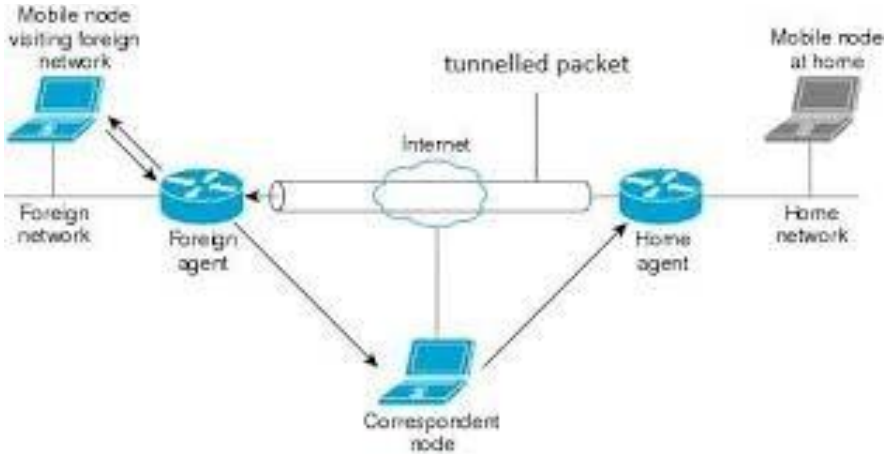
1. **Mobile IP** allows a device to **move across networks** while keeping the **same IP address**.

Key entities

- Mobile Node (MN)
- Home Agent (HA)
- Foreign Agent (FA) (IPv4)
- Care-of Address (CoA)

How it works

1. MN moves to a foreign network
2. Gets a CoA
3. HA tunnels packets to CoA
4. FA delivers packets to MN



CH: The Correspondent Host is the node that communicates with the Mobile Node and sends packets to its home address without being aware of the node's mobility

2. Locator / Identifier Split Separates:

- **Identifier** → *Who you are* (device identity)
- **Locator** → *Where you are* (network location)

LISP (Locator/ID Separation Protocol)

- EID (Endpoint ID)
- RLOC (Routing Locator)

3. Ad-hoc Networking

A **self-configuring network** with **no fixed infrastructure**.

Characteristics

- Nodes act as routers

- Dynamic topology
- Multi-hop communication

Applications

- Disaster recovery
- Military networks
- Sensor networks

4. Routing in Ad-hoc Networks

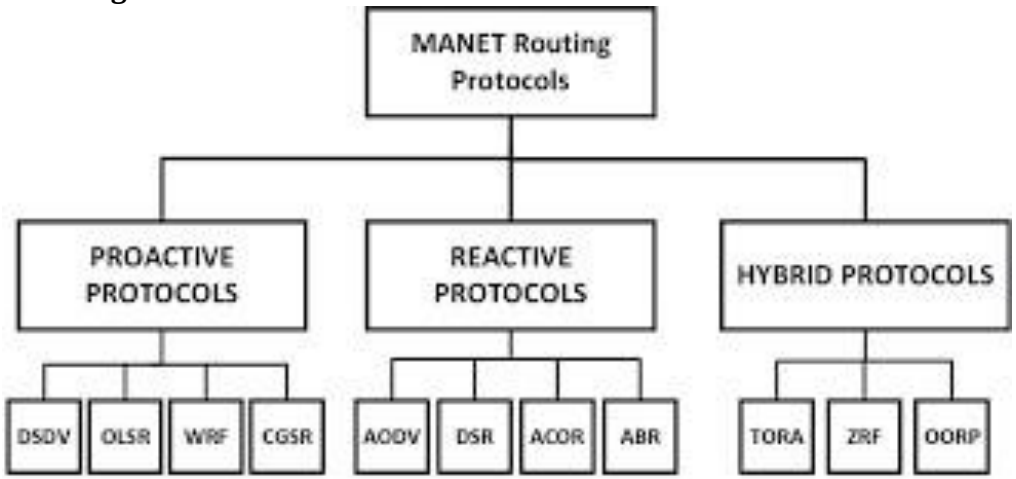


Table 1

Protocol Property	*DSDV*	*DSR*	*AODV*	*OLSR*
Multicast Routes	No	Yes	No	Yes
Distributed	Yes	Yes	Yes	Yes
Unidirectional Link Support	No	Yes	No	Yes
Multicast	No	No	Yes	Yes
Periodic Broadcast	Yes	No	Yes	Yes
QoS Support	No	No	No	Yes
Routes Maintained in	Route Table	Route Cache	Route table	Route table
Reactive	No	Yes	Yes	No

Types of routing protocols

Proactive (table-driven)

- Maintain routes continuously
- Example: **OLSR**

Reactive (on-demand)

- Discover routes when needed
- Examples: **AODV, DSR**

Hybrid

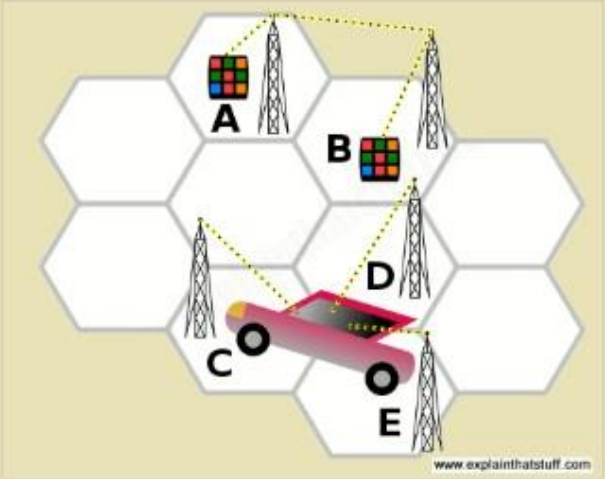
- Mix of both
- Example: **ZRP**

IoT protocol stack

- **Application:** MQTT, CoAP, HTTP
- **Transport:** UDP, TCP
- **Network:** IPv6, 6LoWPAN
- **Link:** IEEE 802.15.4

How cellphone cells handle calls

This picture shows two ways in which cells work.



Simple call

If a phone in cell A calls a phone in cell B, the call doesn't pass directly between the phones, but from the first phone to mast A and its base station, then to mast B and its base station, and then to the second phone.

Roaming call

Cellphones that are moving between cells (when people are walking along or driving) are regularly sending signals to and from nearby masts so that, at any given time, the cellphone network always knows which mast is closest to which phone.

If a car passenger is making a call and the car drives between cells C, D, and E, the phone call is automatically "handed off" (passed from cell to cell) so the call is not interrupted.

The key to understanding cells is to realize that cellphones and the masts they communicate with are *designed* to send radio waves only over a limited range; that effectively defines the size of the cells. It's also worth pointing out that this picture is a simplification; it's more accurate to say that the masts sit at the intersections of the cells, but it's a little easier to understand things as I've shown them.

1. **Expand GSM**

GSM stands for **Global System for Mobile Communications**.

It is a second-generation (2G) digital cellular standard developed to provide voice and data services.

2. **List GSM Mobile Services**

According to Schiller, GSM services are classified into:

1. **Bearer Services**
   - ○ Data transmission services
   - ○ Provide different data rates (e.g., 9.6 kbps)

2. **Teleservices**
   - ○ Voice telephony
   - ○ Emergency calls
   - ○ SMS (Short Message Service)
   - ○ Fax

3. **Supplementary Services** ○ Call forwarding
   - ○ Call waiting
   - ○ Call barring
   - ○ Caller ID (CLIP/CLIR)
   - ○ Multiparty service

GSM architecture consists of three major subsystems:

(a) Mobile Station (MS)

- Mobile Equipment (ME)
- SIM (Subscriber Identity Module)

(b) Base Station Subsystem (BSS)

- BTS (Base Transceiver Station)
- BSC (Base Station Controller)

(c) Network and Switching Subsystem (NSS)

- MSC (Mobile Switching Center)
- HLR (Home Location Register)

- VLR (Visitor Location Register)
- AuC (Authentication Center)
- EIR (Equipment Identity Register)

(d) Operation and Support Subsystem (OSS)

- OMC (Operation and Maintenance Center)

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4. Define Handover in GSM

Handover is the process of transferring an ongoing call or data session from one cell (BTS) to another without interrupting the communication.
It ensures continuity of service when a user moves from one coverage area to another.

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5. What is Localization?

Localization (Location Management) in GSM is the process of determining and updating the current location of a mobile station in the network.
It includes:

- Location update
- Paging

The mobile informs the network when it moves from one location area to another.

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6. List GSM Security Mechanisms GSM security mechanisms include:

1. **Authentication:** Verifies subscriber identity using AuC
2. **Encryption:** Protects user data over the air interface
3. **Anonymity:** Use of TMSI (Temporary Mobile Subscriber Identity) instead of IMSI
4. **Equipment Validation:** Uses IMEI and EIR

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7. What is SIM?

SIM (Subscriber Identity Module) is a smart card inserted into a GSM mobile device that stores:

- IMSI (International Mobile Subscriber Identity)
- Authentication key (Ki)
- Subscriber-related information

It enables user identification and secure access to the GSM network.

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8. Define Satellite Communication

Satellite communication is a wireless communication system that uses artificial satellites to relay signals between distant earth stations.

The satellite acts as a repeater in space.

These four companies are the primary pioneers and current leaders in **mobile satellite services (MSS)**, providing voice and data connectivity in remote areas where cellular networks fail.

Quick Comparison

Feature	Iridium	Globalstar	Inmarsat	ICO
Orbit Type	Low Earth Orbit (LEO)	Low Earth Orbit (LEO)	Geostationary (GEO)	Medium Earth Orbit (MEO)
Coverage	Truly Global (including Poles)	Near-Global (lacks polar regions)	Near-Global (lacks polar regions)	Never fully operational
Primary Use	Handheld voice, IoT, global aviation	Messaging (SPOT), iPhone SOS	Maritime, aviation, broadband data	Historical/Defunct

Key Entity Details

- **Iridium (IRDM)**:
  - Operates a mesh network of **66 LEO satellites**.
  - Unique for its inter-satellite links, allowing it to function without constant ground station proximity.

○ Listed on **NASDAQ** as Iridium Communications Inc. (IRDM).

○ Uses **QPSK** (Quadrature Phase Shift Keying) digital modulation scheme to transmit data over radio waves ● **Globalstar**:

○ Uses a **bent-pipe** architecture, meaning satellites relay signals directly to local ground stations.

○ Highly visible recently due to its partnership with **Apple**, providing the emergency satellite messaging for iPhone 14 and later models.

○ Recently received a **$1.5 billion investment** from Apple for 20% equity.

○ Utilizes **CDMA ● Inmarsat**:

○ The veteran of the group, established originally for maritime safety.

○ Uses large **GEO satellites** parked 35,786km above the equator.

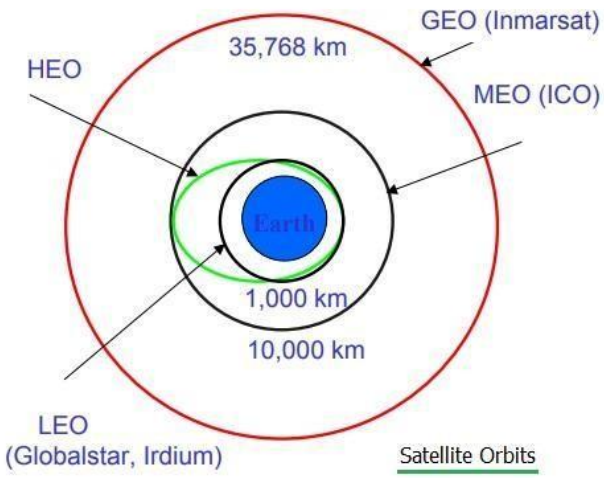
○ Better suited for high-bandwidth needs like Maritime BGAN services.

○ Uses GMSK for specific lower-bandwidth maritime and aeronautical signaling services, as it allows for compact, battery-efficient terminals, GMSK (Gaussian Minimum Shift Keying) is a spectrally efficient digital modulation scheme used to transmit data over radio waves ● **ICO (ICO Global Communications)**:

○ A spin-off of Inmarsat designed to compete with Iridium in the late 1990s.

○ Faced significant financial hurdles and filed for **bankruptcy** in 1999 before its constellation was fully deployed

9. List Types of Satellite Orbits



Main types of satellite orbits:

1. **LEO (Low Earth Orbit)**
    - 500–1500 km altitude
2. **MEO (Medium Earth Orbit)**
    - 5000–20000 km altitude
3. **GEO (Geostationary Earth Orbit)**
    - 35,786 km altitude
    - Appears stationary relative to Earth

10. What is Satellite Handover?

Satellite handover is the transfer of an ongoing call from one satellite beam or satellite to another as the satellite or user moves. It is especially important in LEO and MEO systems due to satellite motion.

1. GSM Protocol Architecture

GSM protocol architecture is organized into **three layers** similar to OSI:

(A) Layer 1 – Physical Layer

- Handles radio transmission
- Channel coding
- Modulation (GMSK)
- TDMA frame structure
    - **Frame Length**: Exactly **90 ms**.
    - **Time Slots**: Each frame is divided into **8 main slots**:
        - **4 Uplink slots**: For transmission from the mobile unit to the satellite.
        - **4 Downlink slots**: For reception by the mobile unit from the satellite.

(B) Layer 2 – Data Link Layer

- LAPDm protocol: **LAPDm** (Link Access Procedure on the Dm-channel) is the **Data Link Layer (Layer 2)** protocol used specifically over the radio interface in GSM-based mobile network
- Error detection and correction
- Reliable signaling transmission **(C) Layer 3 – Network Layer** Divided into three sublayers:

1. **RR (Radio Resource Management)**
    - Channel allocation
    - Handover control
2. **MM (Mobility Management)**
    - Location update

    - Authentication
3. **CM (Connection Management)**
    - Call control
    - SMS handling
    - Supplementary services

2. GSM Call Setup Procedure

Call setup involves signaling between MS, BSS, and MSC.

Steps:

1. **Channel Request** – MS sends request on RACH.
2. **Immediate Assignment** – BSC assigns SDCCH. In GSM, the **BSC (Base Station Controller)** is responsible for assigning the **SDCCH (Standalone Dedicated Control Channel)**
3. **Authentication** – Network verifies subscriber.
4. **Ciphering** – Encryption activated.
5. **Call Setup Message** – Sent to MSC.

The **MSC (Mobile Switching Center)** performs:

- Call switching
- Call routing
- Mobility management
- Interconnection with other networks

It acts like a telephone exchange for mobile networks.

6. **Routing & Alerting** – Called party alerted.
7. **Traffic Channel Assignment** – TCH allocated. (Dedicated Traffic CHannel) 8. **Call Established** – Speech communication begins.

The **RACH** (Random Access Channel) is a common control channel used by a mobile terminal (like a satellite phone) to initiate communication with the network. It is the "first knock" on the door when a device needs to request a dedicated signaling channel for making a call, sending an SMS, or registering its location.

Key Characteristics

- **Direction**: Uplink only (Mobile to Satellite/Base Station).
- **Contention-Based**: It is a shared channel, meaning multiple devices can attempt to send an access request at the same time. This can lead to **collisions**.

- **Protocol**: It typically uses a version of the **Slotted ALOHA** scheme. If a collision occurs, the mobile terminal waits for a random period before trying again.
- **Physical Layer**: The logical RACH is mapped to the **PRACH** (Physical Random Access Channel).

The RACH Procedure (4-Step Process)

When you press "call" on a satellite phone, the following happens:

1. **Msg1 (Random Access Preamble)**: The phone sends a short "preamble" signal on the RACH to the satellite to request resources and allow the network to estimate timing.
2. **Msg2 (Random Access Response)**: The satellite responds on the **AGCH** (Access Grant Channel), providing a temporary ID and a timing advance command to synchronize the phone's clock.
3. **Msg3 (Scheduled Transmission)**: The phone sends a more detailed message (like an RRC Connection Request) using the specific resources granted in Step 2.
4. **Msg4 (Contention Resolution)**: The network confirms which specific device won the access attempt if multiple phones sent Msg1 at the same time

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3. What is Routing in Satellite Systems?

Routing in satellite systems refers to selecting a path for signal transmission between source and destination via satellites and ground stations.

In:

- **Bent-pipe satellites (GEO)** → Satellite only relays signals. Routing done on ground.
- **LEO systems** → May use **inter-satellite links (ISL)**, requiring onboard routing. Routing must consider:
- Satellite movement
- Delay
- Link availability

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4. GSM Authentication Process

Authentication ensures only valid subscribers access the network.

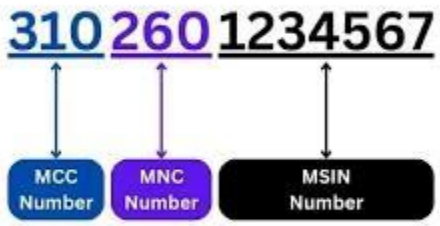
Steps:

1. MS sends **IMSI**.
2. AuC generates:
    - RAND (random number)

    - SRES (Signed Response)
    - Kc (ciphering key)
3. RAND sent to MS.
4. SIM computes SRES using secret key Ki.
5. MS sends SRES back.
6. If matched → authentication successful.
7. Ciphering enabled using Kc. Purpose:

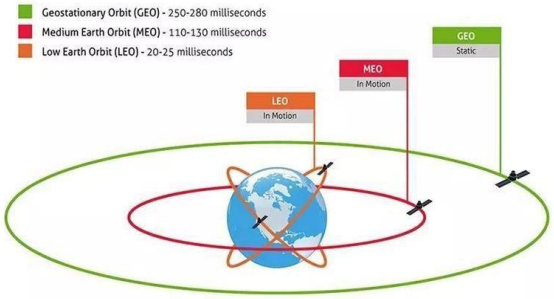
- Prevent unauthorized access
- Protect subscriber identity

IMSI stands for International Mobile Subscriber Identity. It is a unique 15-digit number stored on a SIM card that identifies a specific user on a cellular network (GSM, UMTS, LTE). The IMSI is used by network operators to identify, verify, and authenticate subscribers in the home location register (HLR) or visitor location register (VLR).



Structure: It consists of three parts: Mobile Country Code (MCC, 3 digits), Mobile Network Code (MNC, 2-3 digits), and Mobile Subscription Identification Number (MSIN, 9-10 digits)

5. Compare GEO and LEO Satellites



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Feature	GEO	LEO
Altitude	35,786 km	500–1500 km
Movement	Appears stationary	Moves rapidly
Delay	High (~250 ms one way)	Low
Coverage	Large area	Small footprint
Handover	Not required	Frequent handover
Satellites Needed	3 for global	Many required

6. GSM Localization Procedure

Localization (Location Management) ensures the network knows MS location.

Two main operations:

1. **Location Update**
   - MS updates when entering a new Location Area. ○ HLR and VLR updated.
2. **Paging**

- When a call arrives, network pages MS in the last known Location Area.

Paging is the process the network uses to **find a mobile station (MS)** when there is an incoming call, SMS, or data.

Key Concept:

Network asks: "Where are you?"

Mobile replies: "I'm here!"

2. **Incoming request arrives**
   - A call/SMS comes to the network for a user
3. **HLR is queried**
4. The **Home Location Register (HLR)** provides the current **VLR/MSC** of the user
5. **MSC initiates paging**
   - The **Mobile Switching Center (MSC)** sends a **Paging Request**
6. **Broadcast in Location Area**
   - All **Base Transceiver Stations (BTS)** in that Location Area broadcast the paging message
7. **MS listens and responds**
   - The MS continuously monitors the paging channel
   - When it recognizes its identity (IMSI/TMSI), it sends a **Paging Response**

7. **Connection setup**
   - After response, the network proceeds with call/SMS setup The

process reduces signaling load while ensuring reachability.

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7. Inter-Cell Handover (Analysis)

Source MS → BTS → BSC → MSC → (HLR/VLR) → MSC → BSC → BTS → Destination MS

BTS: Base Transceiver station (Tower that connect mobile device to network) Home Location and Visitor Location Register (HLR/VLR) Inter-cell handover occurs when MS moves to a neighboring cell.

Process:

1. MS measures signal strength & quality.
2. Reports to BSC.
3. BSC decides handover.
4. New channel allocated in the target cell.

5. MS switches frequency/time slot.

Types:

- Intra-BSC ● Inter-BSC
- Inter-MSC Goal:
- Maintain QoS
- Avoid call drop Challenges:
- Timing
- Resource availability
- Signaling overhead

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8. GSM Security Limitations (Evaluation)

Although GSM provides authentication and encryption, limitations include:

1. **One-way authentication**
   - ○ Network authenticates user only.
2. **Weak encryption algorithms** ○ A5/1 and A5/2 vulnerable.
   - ○ Main weaknesses
     - Time–Memory Trade-Off attacks
     - Known-plaintext attacks
     - Precomputed rainbow tables

With tools and precomputed tables, attackers can recover the encryption key **within seconds to minutes**.

3. **No integrity protection**
   - ○ Signaling messages can be modified.
4. **IMSI exposure**
   - ○ If TMSI not assigned. Normally, GSM tries to hide the IMSI using a temporary identifier called Temporary Mobile Subscriber Identity. Normal secure flow:
     1. Mobile phone connects to the network.
     2. The network assigns a TMSI.
     3. The phone uses TMSI instead of IMSI in future communications.
     4. Real subscriber identity stays hidden.

When IMSI gets exposed: If TMSI is not yet assigned, the phone must send its IMSI in plaintext.

This happens in situations such as:

- First time SIM registration
- After phone restart
- Entering a new location area
- When TMSI is lost or expired
- Network requests identity 5.

No end-to-end encryption

- Encryption only on air interface. (only between the mobile phone and the base station, not end to end.)

These limitations led to improved security in 3G (UMTS - Universal Mobile Telecommunications System - end to end encryption).

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9. Satellite Routing Challenges

Satellite routing faces several challenges:

1. **Dynamic topology (LEO/MEO)**
2. **High propagation delay (GEO)**
3. **Limited onboard processing**
4. **Frequent handovers**
5. **Power constraints**
6. **Inter-satellite link management**

Routing algorithms must be adaptive and delay-aware.

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10. Need for New Data Services in GSM (Justification) Original GSM was voice-centric. Growth in:

- Internet access
- Email
- Multimedia
- Mobile commerce

Required enhancements like:

- GPRS (packet switching)
- EDGE (higher data rates) Reasons:

1. Efficient spectrum utilization
2. Always-on connectivity
3. Support for IP-based services
4. Market demand for mobile data

Thus GSM evolved toward 2.5G and later 3G systems.